

IMAGING DIAGNOSIS—HYPEROSTOSIS ASSOCIATED WITH MENINGIOMA IN A DOG

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A 5-year-old neutered male Beagle mix dog had a 5-day history of generalized tonic-clonic seizures. Before the seizures, the dog had a 1–2-month history of progressive right hemiparesis. In computed tomography images, a presumed extraaxial mass with hyperostosis and destruction of the skull covering the mass were identified. Surgical excision was performed and the histopathologic diagnosis was meningioma. Hyperostosis is frequently associated with feline meningioma, but this report documents that hyperostosis may also occur secondary to meningioma in the dog. *Veterinary Radiology & Ultrasound*, Vol. 48, No. 5, 2007, pp 421–423.

Key words: computed tomography, dog, meningioma.

Signalment

FIVE-YEAR-OLD neutered male Beagle mix.

History

The dog was referred to the VCA Aurora Animal Hospital with a 5-day history of generalized tonic-clonic seizures. Before the seizures, the dog had a 1–2-month history of progressive right hemiparesis.

Physical Exam

The dog was demented and nonambulatory. Left head turn was noted along with circling to the left when supported. Postural reactions were absent in the right thoracic and pelvic limbs. Spinal reflexes were normal on all four limbs, and no abnormalities were noted on cranial nerve examination. Based on these signs, the neuro-anatomical localization was a focal left cerebral or diencephalic lesion.

Diagnostic Findings

CSF analysis results were normal white blood cell count (1 cell/hpf; reference range, 0–10 cells/hpf), elevated red blood cell count (260 cell/hpf; reference range 0–10 cells/hpf), low protein (12 mg/dl; reference range 15–35 mg/dl), and normal glucose (65 mg/dl).

Computed Tomography (CT) Imaging Findings

The CT scanner was a helical single-slice unit.*CT imaging of the brain was performed using 3-mm-thick slices from the VCA Aurora Animal Hospital, 2600 W. Galena Blvd., Aurora, IL 60506 (Mercier, Barnes Heller, Bischoff, Looper), and the Antech Diagnostics, Aurora, IL 60506 (Bacmeister).

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contiguous images made before and after intravenous administration of a water-soluble iodinated contrast medium† (370 mg/kg) at a dose of 2.2 ml/kg. There was a large nonuniform enhancing, extraaxial-appearing mass (1.5 × 1.32 × 1.8 cm) in the left cerebral hemisphere, with deviation of the falx cerebri to the right and compression of the left lateral ventricle. The frontal sinuses were asymmetric, with the left being maldeveloped; the lateral compartment of the left frontal sinus is smaller and does not continue as far caudal when compared with the right side. There are more numerous osseous septa and compartmentalization compared with the right frontal sinus. Additionally, there is mild concavity of the inner table of the left frontal bone when compared with the right one. This was thought to be an insignificant congenital malformation. Lysis of the inner table on the left frontal sinus was present caudally. Bone proliferation was present caudal to the left frontal sinus. Contrast enhancement was most severe adjacent to this region of the calvarium with the appearance of subdural and epidural enhancement (Figs. 1 and 2). Considerations included a primary bone tumor, such as chondrosarcoma, osteosarcoma, or multilobular osteosarcoma. The other consideration was a meningioma with secondary hyperostosis.

Diagnosis/Outcome

A craniectomy was performed to obtain a biopsy of the mass and to decompress the brain (Fig. 3). Approximately 80–85% of the volume of the extraaxial mass and osseous component was removed. The patient was discharged from the hospital 3 days after the surgery. At the time of discharge, the patient remained ataxic with absent conscious

*GE CT/e, General Electric Company, Milwaukee, WI.

†Hypaque-76, Amersham Health, Princeton, NJ.



FIG. 1. Transverse computed tomography image of the head. The image is displayed at a soft tissue window ($W=400$, $L=40$). Mild meningeal enhancement (black arrow) is visible with separation from the calvarium by mild subdural/epidural enhancement. This subdural/epidural enhancement was determined to represent the meningeoma at surgery.

proprioception of both right limbs. However, the patient was ambulatory, with assistance, in all four limbs. Despite subsequent radiation therapy and chemotherapy, the dog underwent euthanasia 13 months after surgery due to progressive neuropathy.

Histologically, the mass was composed of spindle-formed cells arranged in whorls and streams. The neoplastic cells had indistinct borders, finely vacuolated to fibrillar eosinophilic cytoplasm, slender ovoid nuclei, vesiculate chromatin, and variably prominent nucleoli. Mitotic figures averaged 4–8 hpf. Collagenous fibrils were interspersed with streams and bundles of neoplastic cells. The histopathologic findings were consistent with a transitional

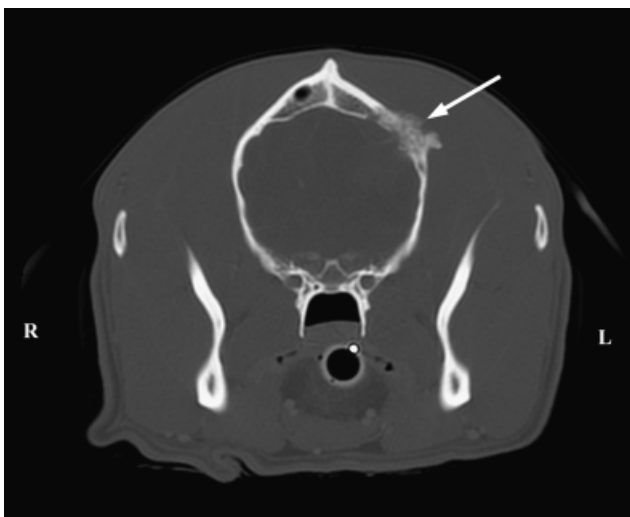


FIG. 2. Transverse computed tomography image of the head. The image is displayed at the bone window ($W=2500$, $L=500$). Notice the localized bony proliferation and mild lysis (white arrow) of the left frontal bone at the caudal extent of the lateral compartment of the left frontal sinus.

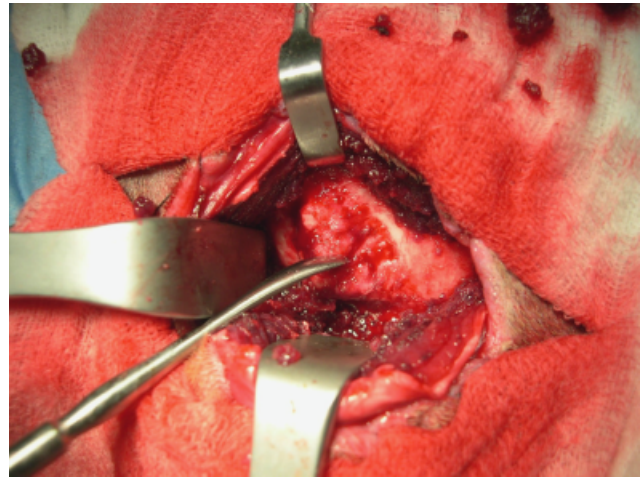


FIG. 3. Photograph acquired during surgery. Note the raised, rough area of skull exposed by retracting the temporalis muscle.

meningioma. Bone fragments were characterized by bone fragmentation with periosteal fibroplasia and new bone formation. Large fragments of skeletal muscle were incorporated into the granulation tissue. There was no evidence of neoplasia on examined sections from the bone and adjacent soft tissues.

Discussion

Tumor-related hyperostosis is bone erosion caused by pressure atrophy with subsequent thickening due to invasion of the Haversian canals by tumor cells.¹ Meningiomas with hyperostosis have been described in cats and humans,^{2,3} but the occurrence in dogs is not well documented.⁴ In this patient, hyperostosis appeared as a focal bony proliferation rather than diffuse thickening of the calvarium as is commonly seen in cats. Although there is anecdotal evidence of this process occurring in dogs, no reports of hyperostosis secondary to meningioma were found.⁴

Meningiomas are the most common primary intracranial tumor in dogs and cats. These mesenchymal tumors most commonly arise from the meningiocytes of the arachnoid layer; however, they could arise from the dura or pia mater. Meningiomas are usually slow growing, benign, solitary tumors and are often adhered to the dura mater.¹ Intracranial and extracranial metastases are rare but have been reported.⁵ There is no particular breed, age, or gender predisposition for canine meningiomas; however, they may have a higher incidence in dolichocephalic breeds.⁶ Older animals are at an increased risk, 95% occurring in dogs older than 7 years.⁶

Surgical resection is the first line of therapy for meningiomas. Tumor location and the infiltrative nature of canine meningiomas can make complete surgical resection difficult. The prognosis for canine meningiomas is guarded due to the high recurrence rate following incomplete re-

section.¹ If complete resection is impossible, radiation therapy is recommended to decrease residual tumor. A median survival time of 7 months was reported with resection alone, compared with a median of 16.5 months with resection, followed by radiation therapy.⁶

Unlike meningiomas in canines, feline meningiomas are usually fibrotic and do not infiltrate adjacent brain tissue.

The mean survival time has been reported to be 485 days after resection of solitary feline meningiomas without radiation therapy.⁷ An association between the presence of hyperostosis and prognosis is unknown.

We document that hyperostosis can occur with canine meningioma and should be considered in the differential diagnosis if such a change is identified.

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